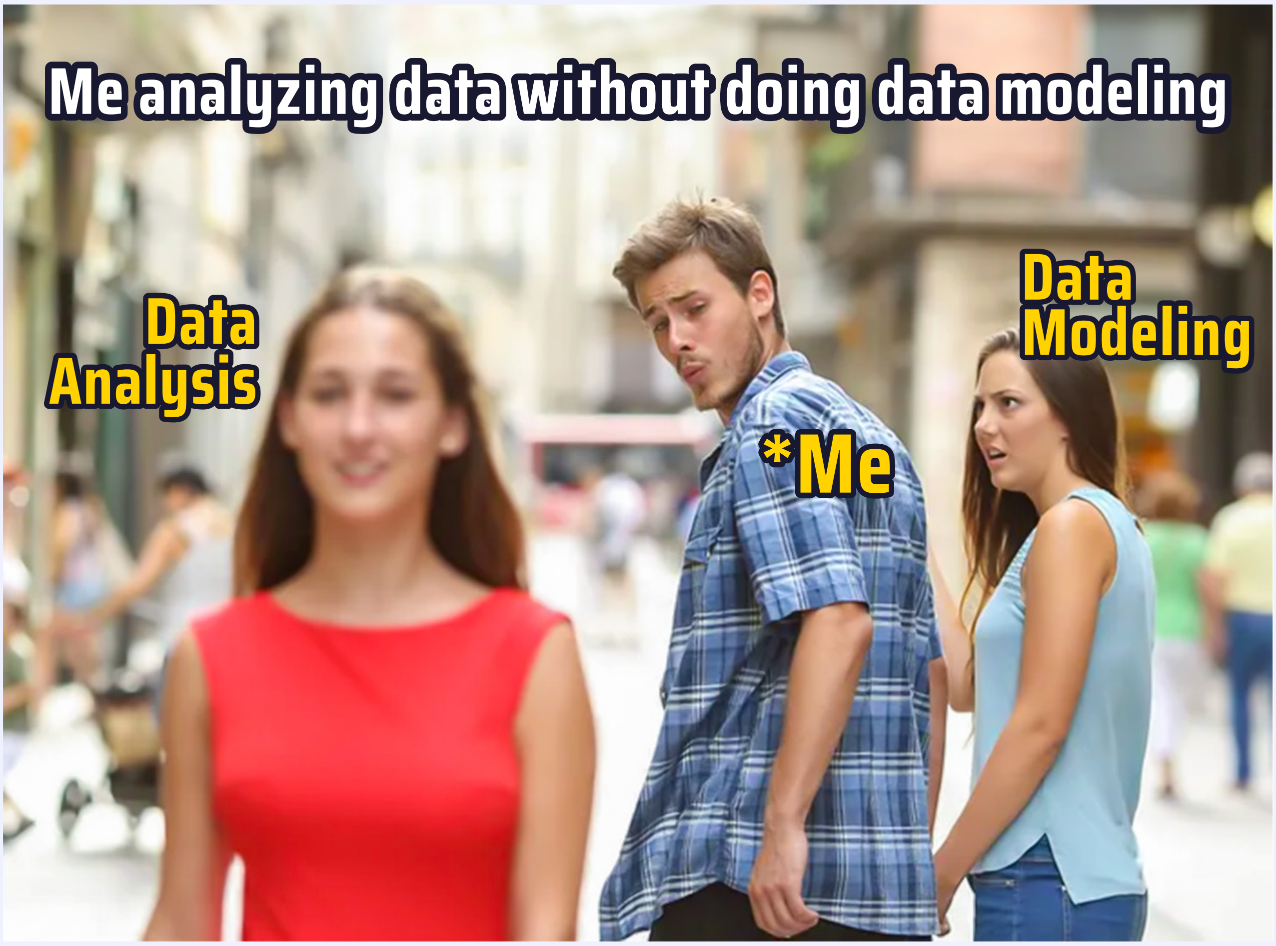
# Data Modeling

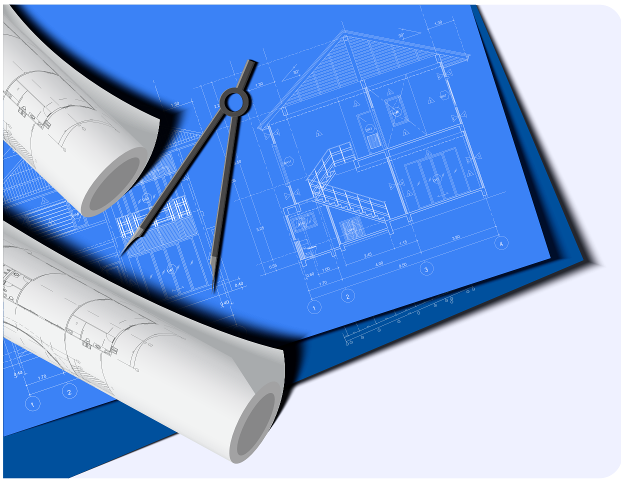
## The Foundation of Data

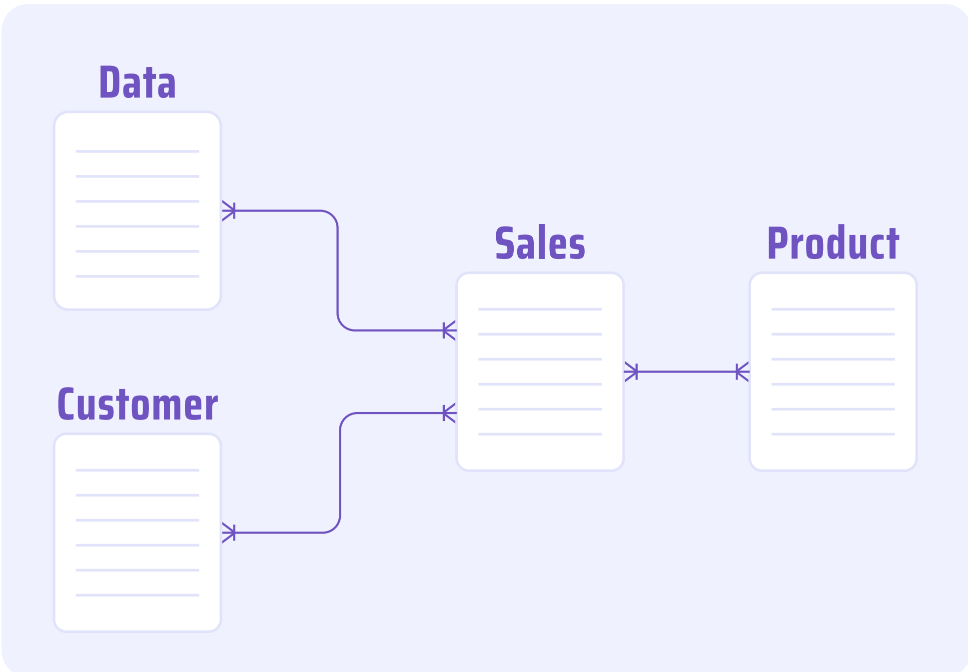



Looking at the meme above, you might understand the importance of data modeling.

Let's dive into the topic 😜

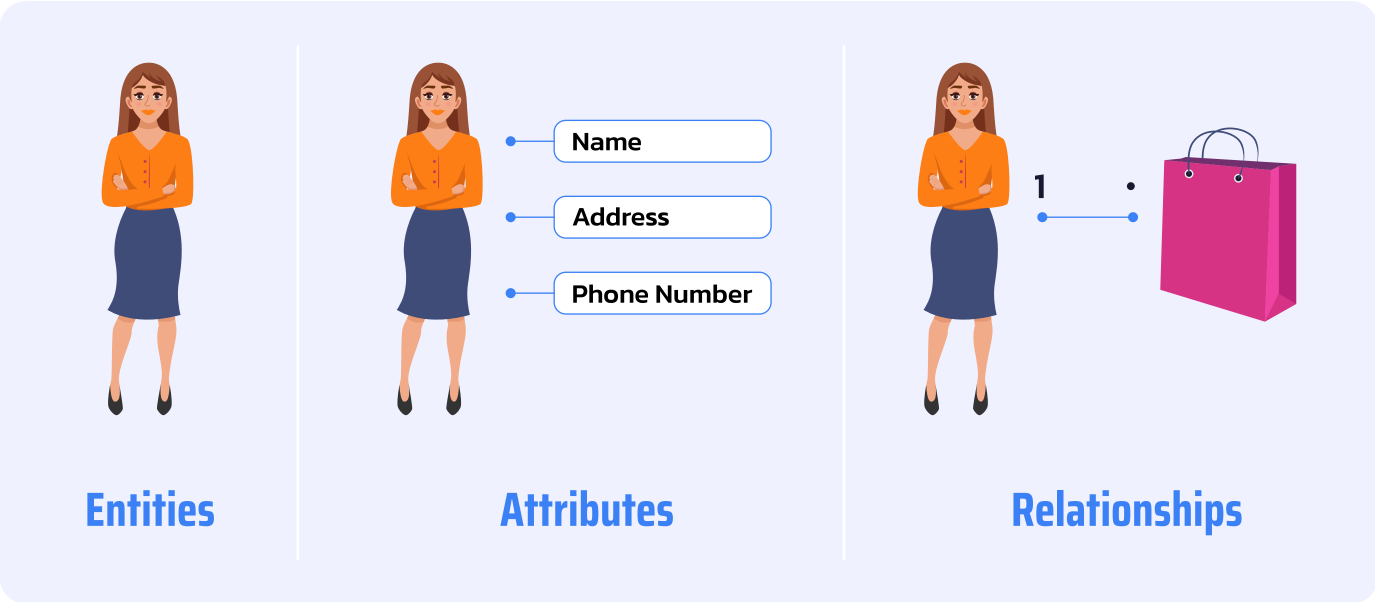
# DATA MODELING:






- **Data modeling is an important step in designing and building a database.**
- Just like a blueprint visualizes the plan and details for a house, data modeling creates a visual representation of data entities and the relationships between data elements.

# COMPONENTS OF DATA MODELING:




## A data model consists of 3 components

### Entities:

These are the main things we want to store information about. For instance, in a business, an entity could be "Customer" or "Product."

### Attributes:

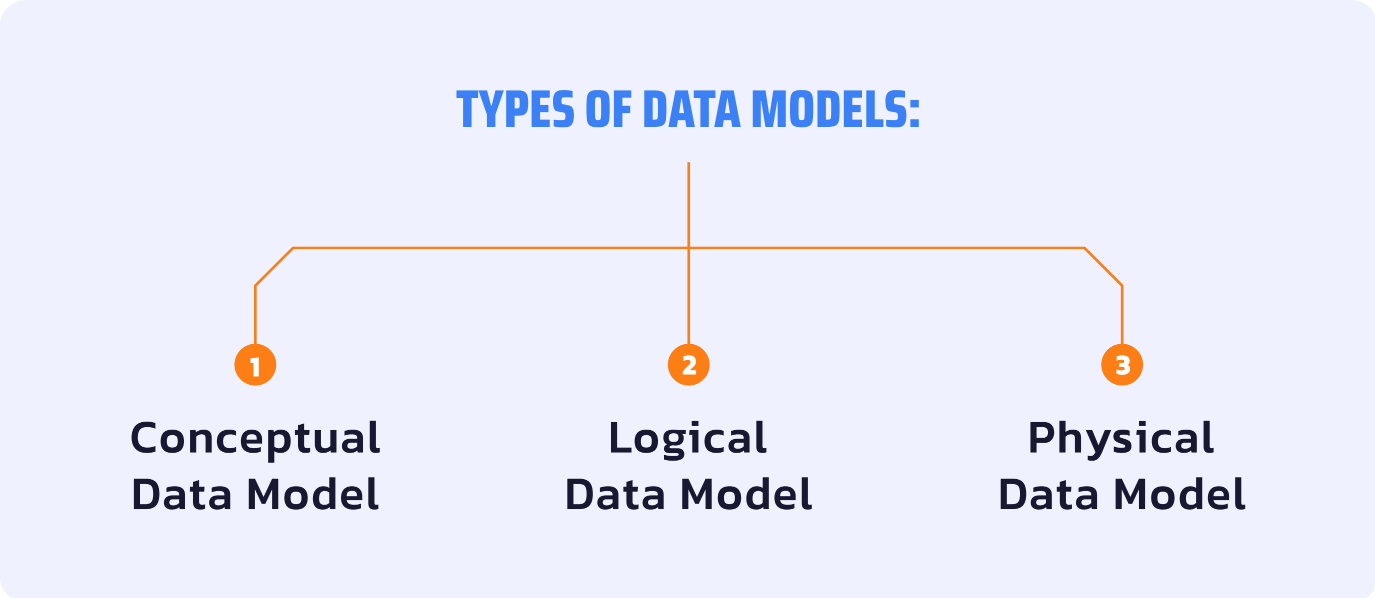
These are the specific pieces of information about an entity. For a "Customer," attributes could include name, address, and phone number.

### Relationships:

These define how entities are connected or related to each other. For example, a "Customer" can have a relationship with an "Order."

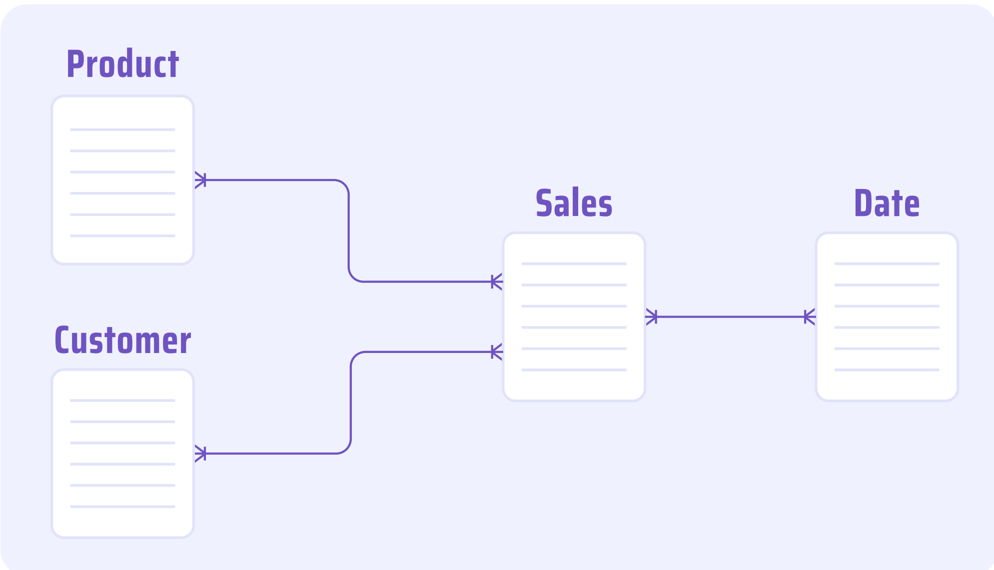
# TYPES OF DATA MODELS:

There are 3 types of data models:
**Conceptual Data Model, Logical Data Model, Physical Data Model**



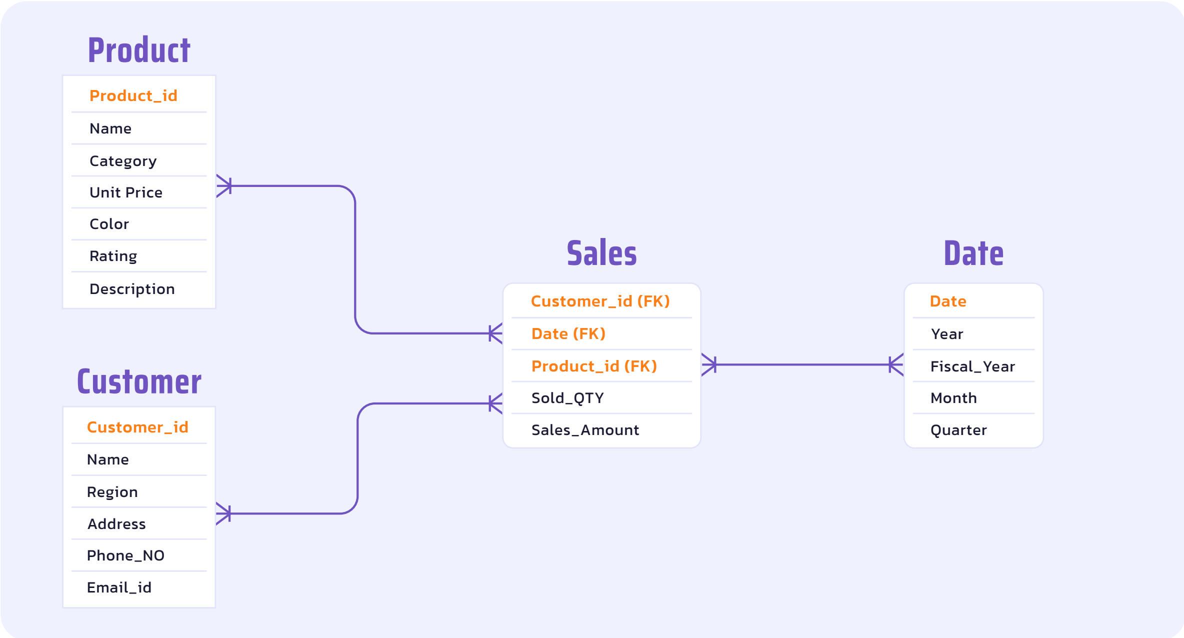

## 1 CONCEPTUAL DATA MODEL

A high-level view of what needs to be stored and how different entities relate to each other. It's like a bird's eye view.



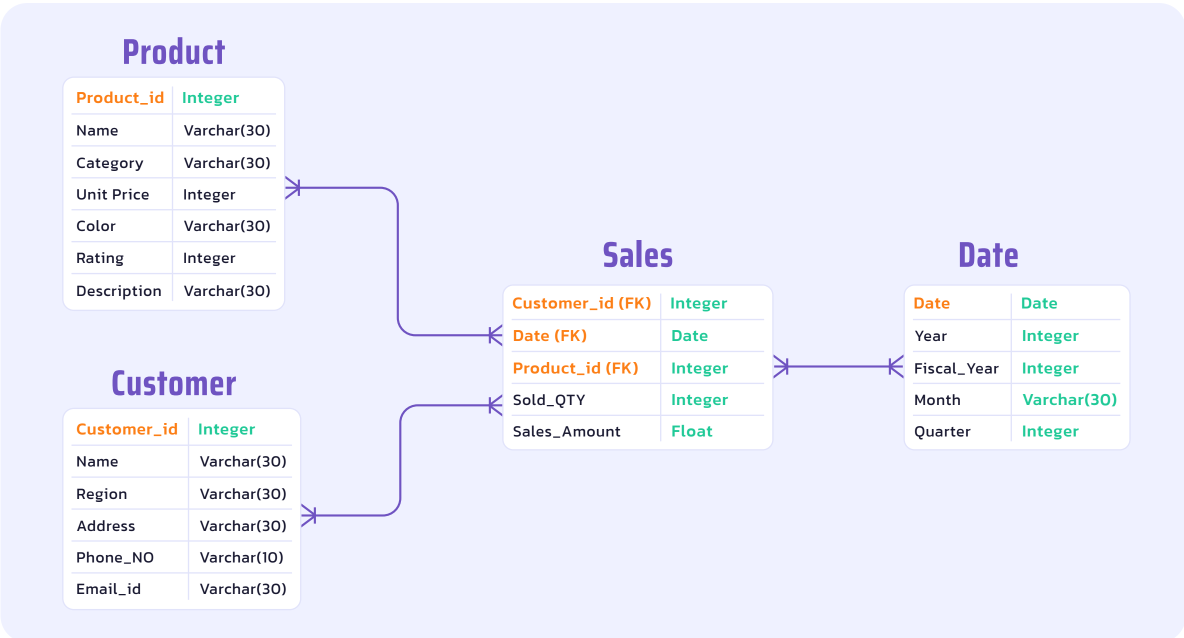
## 2 LOGICAL DATA MODEL

More detailed than the conceptual model, specifying attributes and relationships. It's like a floor plan of a house.




## 3 PHYSICAL DATA MODEL

It specifies how the data will be stored, considering database technologies and constraints. It's like the actual construction of the house.



# TYPES OF TABLES:

Data modeling establishes a connection and flow of data between tables, typically consisting of fact tables surrounded by dimension tables, along with the relationships between these tables.

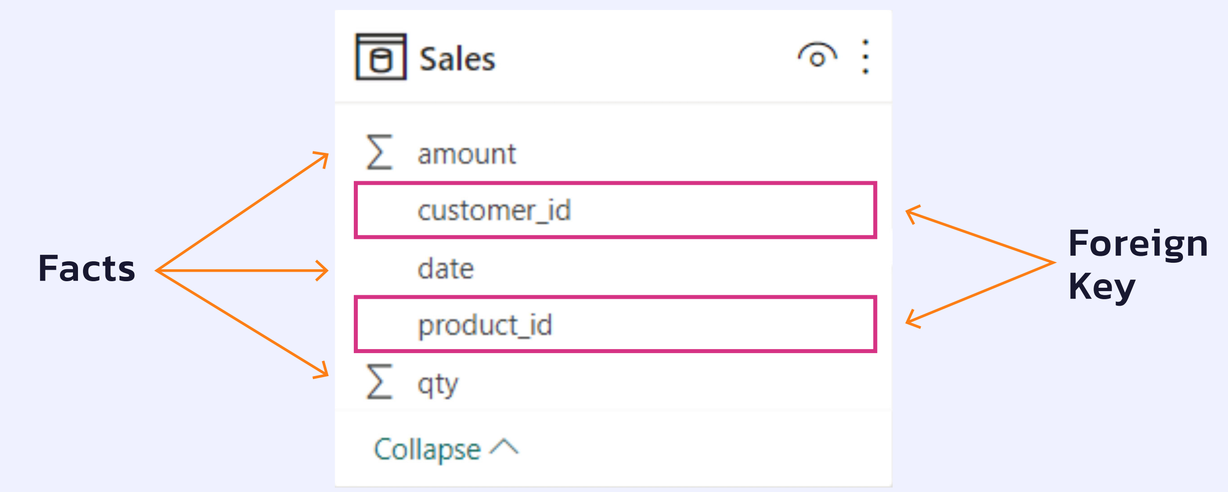
## 1 FACT TABLE:

A fact table contains measurements, metrics, or facts about a business process.

It generally compresses transactional data.

**It has two types of columns:** one representing facts of the business and another containing foreign keys to dimension tables.

**Example:** A Sales fact table contains data on store sales, detailing the quantity of each product sold and the revenue generated from each sale.




Fact Table

**Facts:** amount, date, quantity. These columns represent the business facts.

**Foreign keys:** customer_id, product_id. These columns contain foreign keys that link to dimension tables.

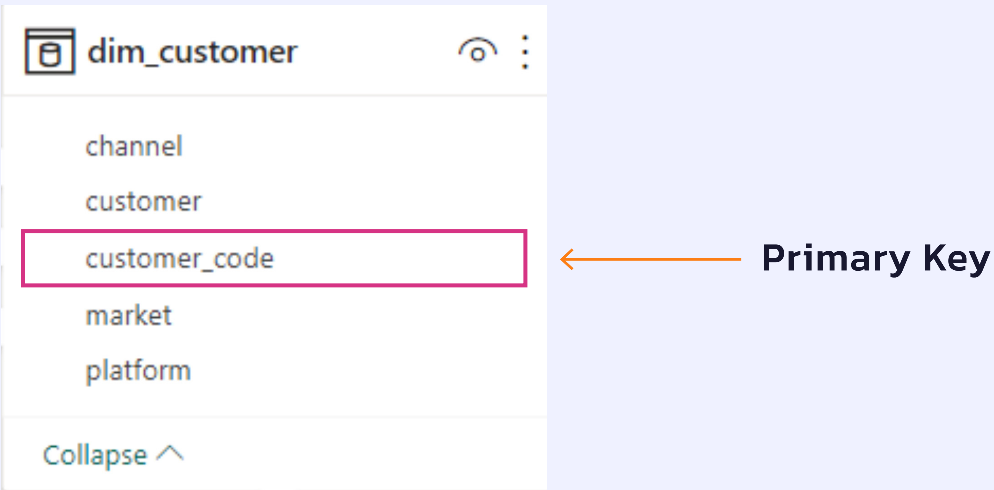
## 2 DIMENSION TABLE:

Fact tables are connected to dimension tables using foreign keys.

Dimension tables consist of attributes that describe the objects of a fact table.

Each dimension table includes a primary key that uniquely identifies each record and using this key dimension table associates with fact tables.

**Example:** A dim_Customer table stores information about the customers who made purchases.




Dimension Table

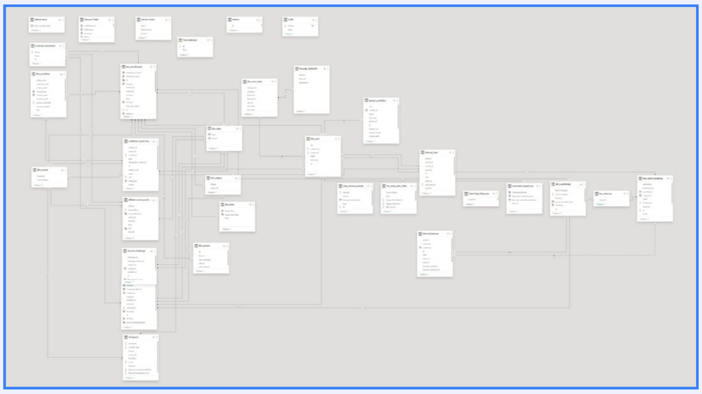
**Primary Key:** Customer_code.

This column contains unique, non-null values associated with records in fact tables.

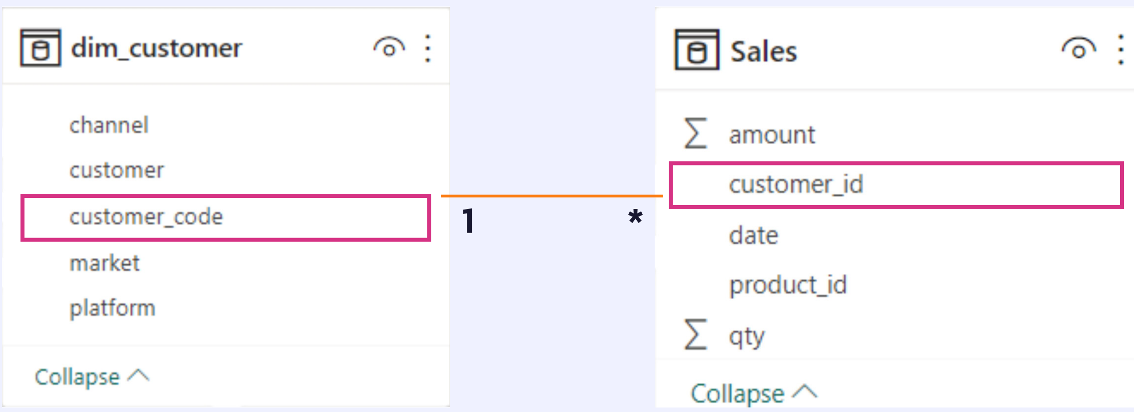
# TYPES OF RELATIONSHIPS:

## Only Data Analysts Can Relate

The Relationship I Want



The Relationship I Get

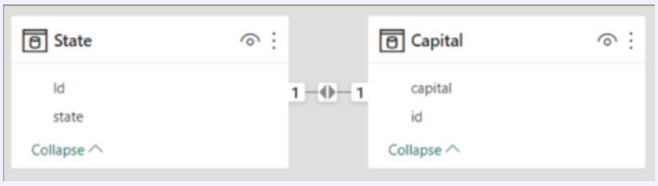



Dimension Table

Fact Table

- Here dim_customers is a dimension table, and Sales is a fact table
- The tables should consist of a common column attribute to make a relationship between tables.
- **There are 4 types of relationships:**
  1. One-to-One relationship
  2. One-to-Many relationship
  3. Many-to-One relationship
  4. Many-to-Many relationship

## 1 One-to-One Relationship:



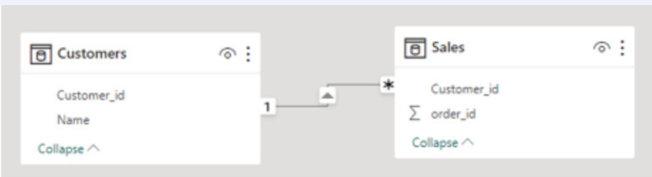

| id | capital |
|---|---|
| 1 | Surat |
| 2 | Chennai |
| 3 | Hyderabad |

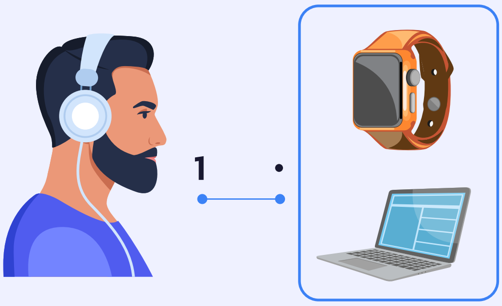
| Id | state |
|---|---|
| 1 | Gujarat |
| 2 | Tamil Nadu |
| 3 | Telangana |


Each row in the first table is mapped to only one row in the second table.

For example, each state has only one capital.

## 2 One-to-Many Relationship:



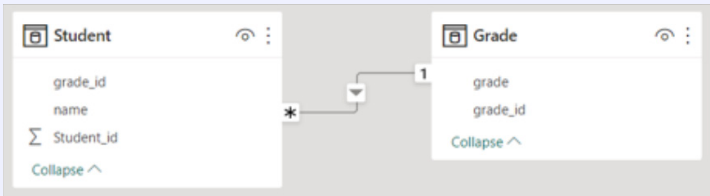

| Customer_id | Name |
|---|---|
| 1 | Dhaval |
| 2 | Hem |
| 3 | Kiran |

| Customer_id | order_id |
|---|---|
| 1 | 102 |
| 1 | 104 |
| 1 | 106 |

In a one-to-many relationship, each row in the first table can be associated with multiple rows in the second table.

For example, a customer can place several orders over time, but each order is tied to a specific customer.

## 3 Many-to-One Relationship:

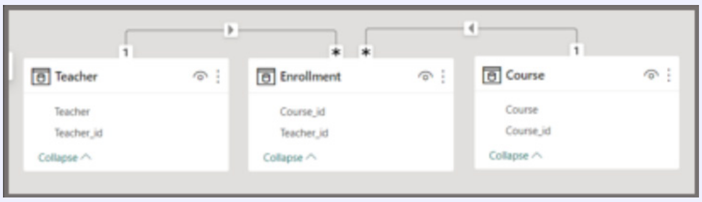

| Student_id | name | grade_id |
|---|---|---|
| 1 | Hemanand | 1 |
| 2 | Dhaval | 1 |

| grade_id | grade |
|---|---|
| 1 | O |
| 2 | A+ |
| 3 | A |
| 4 | B |

Multiple rows in the first table are mapped to a single related row in the second table.

For instance, different students can receive the same grade.

## 4 Many-to-Many Relationship:




| Teacher_id | Teacher |
|---|---|
| 1 | Dhaval |
| 2 | Hem |

| Teacher_id | Course_id |
|---|---|
| 1 | 1 |
| 1 | 2 |
| 2 | 2 |
| 2 | 3 |

| Course_id | Course |
|---|---|
| 1 | ML |
| 2 | Power BI |
| 3 | Excel |

Multiple records from one table relate to multiple records from another table.

## Many-to-Many Relationship:

Dhaval teaches
ML, Power BI

Power BI taught by
Dhaval and Hem

For example, more than one teacher can teach multiple courses. Dhaval teaches ML and Power BI; Power BI is taught by both Dhaval and Hem.